that which in physics is called spacetime: the set of all past and future events, but also those that are ‘neither-past-nor-future’; these do not form a single instant: they have themselves a duration.

In the Andromeda Galaxy, the duration of this expanded present is (with respect to us) 2 million years. Everything that happens during these 2 million years is neither past nor future with respect to ourselves. If a friendly advanced Andromeda civilization decided to send a fleet of spacecraft to visit us, it would make no sense to ask whether ‘now’ the fleet has already left, or not yet. The only meaningful question is when we receive the first signal from the fleet: from that moment on – not earlier – the departure of the fleet is in our past.

The discovery of the structure of spacetime made by the young Einstein in 1905 has concrete consequences. The fact that space and time are intimately connected, as in figure 3.2, implies a subtle restructuring of Newton’s mechanics, which Einstein rapidly completes in 1905 and 1906. A first result of this restructuring is that, as space and time fuse together in a single concept of spacetime, so the electric field and the magnetic fields fuse together in the same way, merging into a single entity which today we call the electromagnetic field. The complicated equations written by Maxwell for the two fields become simple when written in this new language.

There is another implication of the theory, freighted with heavy consequences. The concepts of ‘energy’ and ‘mass’ become combined in the same way as time and space, and electric and magnetic fields, are fused together in the new mechanics. Before 1905 two general principles appeared certain: conservation of mass, and conservation of energy. The first had been extensively verified by chemists: mass never changes in a chemical reaction. The second – conservation of energy – followed directly from Newton’s equations and was considered one of the most *incontrovertible* laws. But Einstein realizes that energy and mass are two facets of the same entity, just as the electric and magnetic fields are two facets of the same field, and as space and time are two facets of the one thing: spacetime. This implies that mass, by itself, is not conserved; and energy – as it was conceived at the time – is not independently conserved either. One may be transformed into the other: only one single law of conservation exists, not two. What is conserved is the sum of mass and energy, not each separately. Processes must exist that transform energy into mass, or mass into energy.

A rapid calculation teaches Einstein how much energy is obtained by transforming one gram of mass. The result is the celebrated formula $E = mc^2$. Since the speed of light c is a very large number, and c^2 an even greater number, the energy obtained from transforming one gram of mass is enormous; it is the energy of millions of bombs exploding at the same time – enough energy to illuminate a city and power the industries of a country for